

Linux Security & Scheduling – Practical Assignments

Topics Covered: Permissions, ACL, Special Permissions, chattr, and Scheduling

Assignment 1: Secure Department Shared Directory

1. Create the following groups: developers, testers, and managers.
2. Create users dev01, dev02, test01, test02, and manager01 and assign them to appropriate groups.
3. Create the directory /company/projects.
4. Configure permissions so developers have read, write, and execute access; testers have read and execute access; and managers have full access.
5. Use ACLs to enforce the required access for the users/groups.
6. Configure a default ACL so that newly created files and directories inherit the required permissions.
7. Apply SGID to /company/projects so that newly created files inherit the directory's group ownership.
8. Apply the sticky bit where appropriate to prevent users from deleting files belonging to other users.
9. Create the file /company/projects/project-plan.txt.
10. Use chattr to protect project-plan.txt from accidental modification or deletion.
11. Verify the file attributes using lsattr and verify permissions using ls -ld and getfacl.
12. Create a scheduled job that performs a daily backup of /company/projects to /backup/projects/.
13. Demonstrate and verify the scheduled backup.

Assignment 2: Secure Application Configuration & Automated Backup

14. Create the users appadmin, appuser, backupuser, and developer.
15. Create the groups appadmins, appusers, and backupadmins and assign users appropriately.
16. Create the application directory /opt/myapp with the subdirectories config, logs, data, and scripts.
17. Configure access so appadmin has full access; appuser has appropriate application access; developer has read access where required; and backupuser has backup-related read access.
18. Use ACLs to implement the access requirements rather than relying only on traditional owner/group/other permissions.
19. Configure default ACLs where appropriate.
20. Determine where SUID, SGID, and sticky bit should or should not be used and implement the appropriate special permissions.
21. Create /opt/myapp/config/application.conf and protect it using an appropriate chattr attribute.
22. Verify the file attribute using lsattr and test modification, deletion, and rename operations.

23. Create an automated backup that runs every day at 11:00 PM and compresses `/opt/myapp/config`.
24. Store the backup under `/backup/myapp/` using the filename format `config-YYYY-MM-DD.tar.gz`.
25. Configure the backup process to delete backups older than 7 days.
26. Verify the permissions, ACLs, file attributes, and scheduled backup.

Assignment 3: Linux Server Security & Compliance Automation

27. Create the users `security01`, `auditor01`, `operator01`, and `admin01`.
28. Create the groups `security`, `auditors`, and `operators` and assign users appropriately.
29. Create the directory structure `/security/confidential`, `/security/reports`, `/security/audit`, and `/security/scripts`.
30. Configure `/security/confidential` so `admin01` and `security01` have full access, `auditor01` has read-only access, and `operator01` has no access.
31. Implement the confidential directory access policy using ACLs and verify it using `getfacl`.
32. Configure `/security/audit` so multiple users can create files, users cannot delete files belonging to other users, and newly created files inherit the appropriate group.
33. Use the appropriate combination of SGID, sticky bit, and ACL and document the reason for each.
34. Create `/security/confidential/security-policy.txt` and protect it using `chattr`.
35. Verify the attribute using `lsattr` and test whether modification, deletion, and rename operations are permitted.
36. Create `/usr/local/bin/security_check.sh` to check permissions, ACL configuration, special permissions, `chattr` attributes, and ownership of important files.
37. Configure the script to write its results to `/var/log/security_check.log`.
38. Schedule the security compliance check to run every day at 01:00 AM.
39. Ensure the log records the date, hostname, directory permissions, ACL status, special permissions, and protected files.
40. Demonstrate the complete security configuration and scheduled compliance check.